MILITARY SPECIFICATION

LUBRICATING AND SERVICING UNIT, AIR OPERATED HIGH AND LOW PRESSURE DELIVERY

This specification is approved for use by the Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of air operated, lubricating and servicing unit for depensing general purpose lubricants, gear and engine oils and chassis grease.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-P-101	Padlock
TT-E-489	Enamel, Alkyd, Gloss (For Exterior And Interior Surfaces)
TT-P-636	Primer Coating, Alkyd, Wood And Ferrous Metal
GGG-O-591	Oiler, Hand
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping Fiberboard
PPP-B-1055	Barrier Material, Waterproofed, Flexible
PPP-T-60	Tape, Packaging, Waterproof
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	Preservation, Methods Of
MIL-B-121	Barrier Material, Greaseproofed. Waterproofed, Flexible
MIL-P-130	Paper, Wrapping, Laminated And Creped
MIL-T-704	Treatment And Painting Of Material
MIL-L-2105	Lubricating Oil, Gear, Multi Purpose
MIL-F-3541	Fitting, Lubrication
MIL-G-3859	Grease Gun, Hand-operated, Lever, Push And Screw Type.
MIL-H-3868	Hose Assemblies, Grease Gun, High And Low Pressure.
MIL-C-4109	Coupling Assembly, Low Pressure, Air-Hose, Quick-Disconnect
MIL-L-4387	Lubrication Dispensing Equipment Accessories.
MIL-H-5593	Hose, Aircraft, Low Pressure, Flexible
MIL-C-7474	Caster, Industrial
MIL-G-10924	Grease, Automotive And Artillery
MIL-V-13811	Varnish, Waterproofing, Electrical, Ignition.
MIL-C-22587	Cartridge, Grease, 14 Ounce (For Cartridge-type Grease Gun)
MIL-L-46167	Lubricating Oil, Internal Combustion Engine, Arctic
MIL-F-82042	Filter And Oiler, Airline

STANDARDS

FEDERAL

FED-STD-H28	Screw Thread Standard For Federal Services
FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Standards And Specifications Order Of Precedence For The Selection Of
MIL-STD-889	Dissimilar, Metals
MIL-STD-1188	Industrial Packing Of Supplies And Equipment
MS24205	Coupling, Grease Gun; Flush Type Nozzle

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

*2.2 Other Publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

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OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 10016, New York.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules And Container Specifications.

(Application for copies of the above publication should be addressed to the American Trucking Association, 1424 16th Street NW, Washington, DC.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Component parts. The lubricating and servicing unit shall consist of the following.

Item Number	Quantity	Description	Requirements
1	1	Cabinet	3.7
2	9	Accessories	3.8
3	3	Pump, Lubricant	3.9
4	1	Reel, Hose	3.10
5	3	Hose, Lubricant	3.11
6	1	Master Valve Assembly	3.12
7	1	Manifold	3.13
8	3	Valve Assemblies	3.14
9	4	Hose, Air	3.15
10	1	Filter	3.16
11	1	Oiler	3.16

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Metals. All metals used in the fabrication of the lubricating and servicing unit shall be of corrosion-resistant type or suitably treated to resist corrosion. Metal subject to deterioration, when exposed to climatic and environmental conditions likely to occur during storage, shipment, or service usage, shall be protected against such deterioration.

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3.4.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5 Design. The design of the lubrication and servicing unit shall follow current commercial practice for this type of equipment. The lubricating and servicing unit shall be designed to satisfactorily dispense chassis lubricant conforming to MIL-G-10924 and oils conforming to MIL-L-46167 and MIL-L-2105 at any temperatures ranging from -20° Fahrenheit (F) to +130°F, in the quantities and under the conditions hereinafter specified.

3.5.1 Air Pressure. All equipment requiring compressed air for operation shall be designed to function at maximum efficiency from air line pressure of between 90 and 175 pounds per square inch (Psi) and withstand stall pressure up to and including 200 psi air line pressure without damage.

3.5.2 Maintenance. The design of the lubricating and servicing unit shall be such that adjustments, repairs or replacement of components can be accomplished by personnel of operating units utilizing general purpose tools.

3.6 Performance. The lubricating and servicing unit shall be capable of meeting the following performance requirements:

a. Operating temperatures ranging from -20°F to +130°F and storage in temperatures ranging from -40°F to +160°F.

b. A relative humidity of 95 to 100 percent at an operating temperature of 130°F and a storage temperature of 160°F.

c. Lubricant displacements utilizing air pressures and ambient temperatures as indicated in Table I.

3.7 Cabinet. The cabinet shall be adequate to contain three 120 pound grease drums, and other components specified herein, and shall be constructed in such a manner as to provide easy accessibility to the interior. The cabinet shall be fabricated of good quality sheet steel or aluminum, adequately reinforced to maintain the drums and components in rigid alignment and shall be securely mounted on the mobile base.

3.7.1 Base. A suitable metal base of sufficient strength to withstand a minimum load of 800 pounds shall be provided. The base may be fabricated of sheet steel or aluminum and shall be properly reinforced, where required,to withstand the strains, jars and other conditions incident to shipping and service. The base shall be mounted on two 8-inch diameter wheels, and on two 6-inch swivel type casters conforming to MIL-C-7474. Each wheel shall have a minimum load carrying capacity of 400 pounds and shall be equipped with roller bearings in the hub and a semi-pneumatic tire with a tread width of not less than 2 inches. The fixed wheels shall be attached to a suitable rigid axle not less then 24 inches long. The base shall also be equipped with a suitable foot brake capable of holding the fully equipped and loaded unit.

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TABLE I. Lubricant displacement.

	MINIMUM DELIVERY PER MINUTE	AIR PRESSURE USED IN PSI	AMBIENT TEMPERATURE IN DEGREES FAHRENHEIT
High Pressure Pump Delivery (Free Run) Using Chassis Grease MIL-G-10924 NOTE: Static pressure developed by the pump shall be not less than 4800 psi (minimum).	20 Ounces 14 Ounces 5 Ounces	120 120 120	+70 ± 5 0 ± 2 -20 ± 2
Low Pressure Pump Delivery (Free Run) Using Lubricating Oil, Gear MIL-L-2105	16 Pounds (SAE 90) 12 Pounds (SAE 90) 8 Pounds (SAE 90)	120 120 120	+70 ± 5 0 ± 2 -20 ± 2
Low Pressure Pump Delivery (Free Run) Using Lubricating Oil, Internal-Combustion Engine MIL-L-46167	3 Gallons 1 3/4 Gallons 1 1/4 Gallons	120 120 120	+70 ± 5 0 ± 2 -20 ± 2

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3.7.2 Compartments. The cabinet shall provide six compartments: one compartment for storing lubricant containers, pumps and drum covers; one compartment of suitable size to receive a chassis grease hose reel; two compartments to receive the gear oil and motor oil hoses, these compartments of suitable size to allow the hose to slide in and out of the compartment easily; one compartment to house the three lubricant control valves, when connected to their respective dispensing hoses; and one compartment for storing hand grease and oil guns and other accessories (see 3.8). Where required, the compartments shall be equipped with access openings through which the lubricant hose will pass. Such access openings shall be equipped with hose rollers with a minimum of four rollers equally spaced in a 360° circle for easy guidance of the hose without binding or pinching.

3.7.2.1 Compartment doors. Suitable doors or panels with snap-lock fasteners shall be provided for access to lubricant control valve assemblies, for access to lubricant pumps, for access to hand guns (grease and oil), and for removal and replacement of lubricant containers.

3.7.3 Lifting and tie-down devices. Four suitable lifting and tie-down devices shall be attached to the cabinet base to facilitate the use of cargo slings for loading and unloading and for servicing the unit during air, water, rail or motor transport. Such devices shall be of sufficient strength to withstand any handling condition such as rapid lowering and violent braking of load by crane.

3.7.4 Cabinet handle. A suitable pulling-pushing handle shall be attached to the caster end of the cabinet. The cabinet shall be suitably reinforced at the handle attaching points to prevent distortion of the handle and cabinet.

3.8 Accessories. Unless otherwise specified (see 6.2), the following accessories shall be furnished with each lubricating and servicing unit. When specified (see 6.2), a hand operated grease gun conforming to MIL-G-3859 shall be furnished. An initial supply of six (6) grease cartridges conforming to MIL-C-22587 shall be furnished with the MIL-G-3859 grease gun. All grease gun equipment shall be furnished, where practical, with quick change features permitting the equipment to be changed without the use of tools.

Quantity	Item
1 each	Padlock with 2 keys, conforming to FF-P-101, type EPA, size 2 inch, laminated steel case.
1 each	Grease Gun, lever operated, 15 ounce (oz) capacity conforming to MIL-G-3859.
1 each	Grease Gun, high pressure, 3,500 to 10,000 psi 2 1/2 oz capacity, conforming to MIL-G-3859.
1 each	Oil Gun, suction type, 11 oz capacity, type II, class C, style 4, conforming to GGG-O-591.

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Quantity	Item
1 each	Flexible extension, type IV, class I, style B, conforming to MIL-L-4387.
1 each	Coupler, shafer type fitting, conforming to MS24205.
1 each	Oiler, steel round, pump feed, 1/2 pint capacity, 6 inch rigid bent spout, conforming to GGG-O-591 type II, class B.
1 each	Oil spray Gun, air operated, on quart capacity with connector type II, class A, style I, and coupling type I, class A, style I, conforming to MIL-C-4109.
1 each	Adapter-Coupler, giant button-head, with 1/2 inch National Taper Pipe (NPT) female thread.

3.9 Pumps, lubricant.

3.9.1 General. All lubricant pumps shall be designed for pumping grease and oil to within a maximum of two inches from the bottom of the lubricant containers. The pump shall not consume more than 13 cubic feet per minute (cfm) of free air delivered at 120 psi pump gauge pressure when pumping the grease or oil hereinafter specified at free lubricant flow, through dispensing hoses and control valves at an ambient temperature of 70°F. Pumps shall be capable of performance, environmental and storage requirements specified in 3.6.

3.9.2 Pumps, low pressure, gear and engine oils. Pumps shall be double acting, self priming, air motor driven with maximum pressure ratios of not more than 6 to 1 for dispensing arctic gear oils or 4 to 1 for dispensing engine oils conforming to MIL-L-2105 and MIL-L-46167, respectively. Pumps shall be equipped with an air pressure gauge (0 to 200 psi scale), air pressure regulators and ball type foot valves. With the air gauge on the pump registering 120 psi, pumps shall be capable of delivering gear or engine oils from 120 pound lubricant drums in the quantities and at ambient temperatures specified in Table I. Suitable drum covers shall be provided to support the motor and gear oil pump to prevent dirt and foreign matter from entering the container. The covers shall incorportate a seal and at least 3 wing bolts, or the equivalent, for attaching to the drum. The cover for the motor oil cantainer shall be provided with a 4-inch refill opening with spring sliding cover.

3.9.3 Pump, high pressure, chassis grease. The pump shall be double acting, self priming, air motor driven with a minimum pressure ratio of 40 to 1 for dispensing grease conforming to MIL-G-10924 from standard 120 pound open top drums. Pump shall be equipped with an air pressure gauge (0 to 200 psi scale), and an air pressure regulator. With the air gauge on the pump registering 120 psi, the pump shall be capable of delivering grease under a minimum pressure of 4800 psi in the quantities and at ambient temperatures specified in Table I. A suitable drum cover and grease follower plate, for use in a standard 120 pound open top drum, shall be provided. The cover shall be designed to support the pump and incorporate a seal and have at least 3 wing bolts, or the equivalent, for attaching to the drum.

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3.9.4 Pump screen. The suction end of the gear and engine oil pumps and the chassis grease pump shall be fitted with a 30 mesh wire screen or other filtering device to prevent entrance of foreign matter into the pump. When a screen is furnished, it shall be made of corrosion-resisting wire or wire with a corrosion-resistant plating. The wire shall be not more than 0.014 inch diameter and width opening between wires shall not be more than 0.021 inch. The screen or filtering device shall be readily removable, and it shall not be necessary to disassemble any part of the pump for cleaning purposes. The screen or filtering device shall not reduce the output of the pump.

3.10 Reel, chassis lubricant. A suitable lubricant hose reel shall be provided and mounted within the cabinet so that the hose is pulled out of the end of the cabinet adjacent to the engine oil and gear oil hose and control valves. The reel shall be of the spring actuated hose retraction type with a spool of ample capacity to accommodate 30 feet of 1/4 inch inside diameter (id) lubricant hose without pinching or binding in either hose pull out or retraction. The reel shall be provided with a latching or stop mechanism to automatically lock in a minimum of two stations in any one full revolution of the reel. The reel shaft and connections thereto shall have adequate packing to prevent lubricant leakage. "Anchor" elbow unions, (pin type or swivel nut), shall be provided on the inlet side of the reel shaft and discharge side of the reel to facilitate its removal. The reel shall be furnished less the reel case.

3.11 Hose. All lubricant hoses shall be provided in sizes and length with end fittings as hereinafter specified for each type, style and grade hose.

3.11.1 High pressure. The hose assembly for dispensing chassis grease which is stowed on the hose reel shall be 1/4 inch id, having a minimum extendable length of 25 feet with 1/4 inch external and internal NPT end fittings conforming to type I, style 3, grade B, MIL-H-3868.

3.11.2 Low pressure. The hose assemblies for dispensing engine and gear oils shall be either 3/8 or 1/2 inch id, having a minimum extendable length outside the cabinet of 13 feet with NPT end fittings conforming to type II, style 3, grade B, MIL-H-3868.

3.12 Master valve assembly. A master air line valve with quick-disconnect coupler conforming to MIL-C-4109 that will relieve the air and lubricant pressure in the lines when the unit is not in use shall be provided.

3.13 Manifold. A suitable air manifold shall be provided to facilitate connection of air lines to the lubricant pumps. It shall terminate at the exterior of the cabinet with a type I quick-disconnect coupling conforming to MIL-C-4109.

3.14 Control valve assemblies. High and low pressure lubricant control valve assemblies shall be provided complete with accessories (hose to control valve) as hereinafter specified for each type, class and style assembly.

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3.14.1 Control valve assembly, high-pressure, chassis grease. A type VI, class 1, style A, control valve assembly equipped with a type VII, class 3 swivel and an adapter, hydraulic coupler, type IV, class 1, style A, conforming to MIL-L-4387 shall be furnished for dispensing chassis grease, all with threads to fit the MIL-H-3868 hose furnished. The hydraulic coupler shall fit grease fittings of the hydraulic type conforming to MIL-F-3541. A suitable pressure reducing device shall be provided, in or with the high pressure control valve of such nature that when at the low pressure position, the chassis grease pump pressure ratio is reduced to approximately 8 to 1 of eight times the air pressure supplied to operate the pump, and when placed in the high pressure position the full pump pressure shall be available. Full delivery of the pump shall not be affected by the use of the pressure reducing device.

3.14.2 Control valve assembly, low-pressure, motor oil. A type VI, class 2, style B, control valve equipped with a type VII, class 1 hose swivel conforming to MIL-L-4387 or a control valve similar in all respects to the type VI, class 2, style B valve except that an integrated swivel connection is utilized and in each case with threads to fit the MIL-H-3868 hose furnished, shall be furnished for dispensing and metering motor oil. The totalizing meter shall measure in pints and quarts and fractions thereof.

3.14.3 Control valve assembly, low-pressure, gear oil. A type VI, class 2, style A control valve equipped with a type VII, class 1 hose swivel conforming to MIL-L-4387 or control valve similar in all respects to the type VI, class 2, style A valve except that an integrated swivel connection is utilized, and in each case with threads to fit the MIL-H-3868 hose furnished, shall be for dispensing and metering gear oil. The totalizing meter shall measure in pints and quarts and fractions thereof.

3.15 Air hose assemblies. All air hoses provided shall conform to the requirements for size 4, MIL-H-5593.

3.15.1 Air connection hoses. Suitable lengths of air hose with one end terminating in a 1/4 inch external NPT end fitting and the opposite end terminating in a quick-disconnect coupling comforming to MIL-C-4109 shall be provided for use in the following locations:

a. From air manifold to gear oil pump.
b. From air manifold to motor oil pump.
c. From air manifold to chassis grease pump.

Suitable lengths of air hose with ends terminating with 1/4 inch external NPT end fittings shall be provided for use in connecting the air manifold, air line filter and oiler, and master shut-off valve.

3.15.2 Air source hose assembly. One air hose assembly, not less than 30 feet in length, with ends terminating in quick-disconnect couplings conforming to MIL-C-4109, shall be provided to effect connection of air source to the unit.

3.15.3 Couplings. Suitable air hose quick-disconnect couplings conforming to MIL-C-4109 shall be provided for use in effecting connection of air hoses to lubricant pumps to facilitate replacement of lubricant containers.

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3.16 Filter and oiler. One air line filter and oiler conforming to MIL-F-82042, type II, shall be connected to the main air supply line in such a manner as to supply ample lubrication for the lubricant pumps and exclude foreign matter from entering the air pump. The air line filter and oiler shall operate efficiently under condition specified in 3.5.1.

3.17 Lubrication. Components of the lubricating and servicing unit requiring lubrication shall be provided with suitable closed fittings or cups to prevent foreign matter from entering the bearing.

3.18 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.19 Threads. All threaded parts shall be in accordance with Handbook H-28.

3.20 Treatment and Painting. The lubricating and servicing unit shall be treated and painted in accordance with MIL-T-704, type A-1. The finish color shall be gray, Number 16187, conforming to FED-STD-595 and coating materials shall conform to TT-P-636 and TT-E-489.

3.21 Identification of Product. Equipment, assemblies, and parts shall be permanently and legibly marked with the manufacturer's part number in accordance with MIL-STD-130, except parts which do not have suitable or sufficient surface, commonly known commercial items which present no identification problem (such as nuts, bolts, and washers), and components of sub-assemblies and assemblies which are not normally subject to disassembly or repair.

3.22 Technical publications. Unless otherwise specified (see 6.2) two copies of a handbook of operation and servicing instructions covering the various components and accessories shall be furnished and packaged with the unit, whether manufactured by the contractor or not. Sub-data shall include low temperature starting, operation and shutdown procedures. Cautions and safety practices shall be included throughout the texts. The publications shall be bound together in single cover and in a weatherproof jacket or envelope. A suitable metal jacket shall be attached to the interior of the enclosure to receive the handbooks.

3.23 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with the best commercial practice for this type equipment.

3.23.1 Metal fabrication. Metal used in the fabrication of the lubricating and servicing unit shall be free from kinks and sharp bends. The straightening of material shall be accomplished by methods that will not cause injury to the metal. Shearing and clipping shall be accomplished neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends of major character shall be made with metal dies or fixture to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly

3.23.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers and lock washers shall be

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provided in accordance with the best commercial practice, and all bolts and screws shall be tight.

3.23.3 Rivet connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. When not countersunk or flattened, rivet heads shall be of approved shape and uniform size for the same diameter or rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.4 Fasteners. Intricate locking devices and threaded fasteners which can be over-torqued shall be avoided. Covers, doors or plates which require removal for component adjustment shall be equipped with quick-disconnect fastening. All screws pins, bolts and similar parts shall be installed with an adequate means for preventing loss of proper adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, or staked or otherwise permanently deformed.

3.23.5 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall provide adequate strength in the material connected. Welding shall be in accordance with a welding code acceptable to the procuring activity.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed.

4.2 Classification of tests. The inspection and testing of the lubricating and servicing unit shall be as follows:

a. Preproduction tests (see 4.3)
b. Acceptance tests (see 4.4)

4.3 Preproduction testing.

4.3.1 Preproduction test sample. The preproduction test sample shall consist of complete lubricating and servicing units that the contractor proposes to furnish the Government under this specification. The number of units to be subjected to the preproduction test shall be as designated by the procuring agency (see 6.2). They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.3.2 Preproduction tests. Preproduction tests shall consist of all tests described under 4.5.

4.4 Acceptance tests. Acceptance tests shall consist of individual tests specified herein.

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4.4.1 Individual tests. Each lubricating and servicing unit shall be inspected and tested as specified in 4.5.1 and 4.5.2.

4.5 Examination and tests.

4.5.1 Examination of product. Each unit shall be examined sufficiently to assure compliance with this specification relative to material, proper assembly, completeness, finish, marking and overall quality of workmanship.

4.5.2 Operation. Each unit shall be charged with lubricant as specified in 3.5 and connected to a source of compressed air conforming to 3.5.1 and tested for proper operation of pumps, valves, et cetera. During this test, a check shall be performed to determine if there are any air or lubricant leaks in the system.

4.5.3 The lubricating and servicing unit shall be filled with lubricants specified in 3.5 and tested to verify compliance with 3.6. In addition, lubricant pumps shall be tested to verify conformance to 3.9.

4.5.4 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

*5. PACKAGING.

*5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preserving. Before preservatives are applied metal surfaces of parts involved shall be cleaned and dried in accordance with MIL-P-116. Preservatives shall conform to the applicable specifications listed herein and MIL-P-116. All references to "P" type preservatives are those listed in MIL-P-116.

5.1.1.2 Wheels and casters. The wheels and casters shall be removed from the lubricating unit and the bearings shall be lubricated with the manufacturer's recommended lubricant. Each wheel and caster shall be wrapped in barrier material conforming to MIL-B-121, grade A, type II, class II. Bolts, washers and nuts resulting from removal shall be cleaned by method C-1, coated with type P-1 preservative, and placed in a good quality drawstring bag. The wheels, casters and the bag of bolts, washers and nuts shall be packaged in a fiberboard container conforming to PPP-B-636, class domestic. The box shall be closed and sealed in accordance with the appendix to the box specification. The carton shall be placed in the metal cabinet within the unused space provided for the 120 pound drums. The box shall be adequately secured within the cabinet to prevent shifting and damage to other component parts.

5.1.1.3 Unpainted surfaces. Unpainted metal surfaces of component parts shall be coated with type P-1 preservative. Unpainted information and data

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plates shall be coated with ignition waterproof varnish conforming to MIL-V-13811.

5.1.1.4 Pump assemblies. The engine oil pump, gear oil pump, and the grease pump shall be mounted on the adapter provided, and shall be connected to a standard 120 pound lubricant drum containing type P-10, grade 2, preservative, and the pumps shall be operated for a sufficient length of time to insure complete coverage of all internal surfaces. Excessive preservatives shall be drained and the external surfaces of the pump casings coated with type P-3 preservative.

5.1.1.5 Hose. The hose couplings shall be coated with type P-1 preservative. The chassis grease hose shall be compactly wound on the hose reel and secured to prevent unreeling, and the reel shall be locked. The open end of each hose shall be sealed with pressure-sensitive tape conforming to PPP-T-60.

5.1.1.6 Control valves. Prior to attaching to the applicable hose, the grease control valves, the oil dispensing valve, meters and the gear lubricant dispensing valve shall be coated internally with type P-10, grade 2 preservative.

5.1.1.7 Lubricator. One lubricator, as specified, constitutes one unit. All exterior finished sufaces shall be protected by wrapping with barrier material conforming to MIL-B-130, type I, and cushioned with excelsior pads or flexible corrugated fiberboard, or other suitable materials. Wrapping and cushioning shall be secured with tape conforming to PPP-T-97, type III.

*5.1.2 Commercial. Commercial packaging shall conform to the requirements of MIL-STD-1188.

*5.2 Packing. Packing shall be level A,B,or Commercial, as specified (see 6.2).

*5.2.1 Level A. Each unit, packaged as specified in 5.1, shall be packed in a wood cleated-plywood shipping container conforming to PPP-B-601, overseas type, style and grade optional. The lubricator shall be shrouded with waterproofed paper conforming to PPP-B-1055, Class H-5. The unit shall be adequately blocked and braced to prevent movement. The shipping container shall be of minimum cube and tare consistent with the protection required. Closure and strapping shall be in accordance with the appendix to PPP-B-601.

*5.2.2 Level B. Each unit packaged as specified in 5.1, shall be packed in a wood cleated-plywood shipping container conforming to PPP-B-601 domestic type style and grade optional. The unit shall be adequately blocked and braced to prevent movement. The shipping container shall be of minimum cube and tare consistent with the protection required. Closure and strapping shall be in accordance with the appendix to PPP-B-601.

*5.2.3 Commercial. Commercial packing shall conform to the requirements of MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129, or MIL-STD-1188, as applicable.

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6. NOTES

6.1 Intended use. The lubricating and servicing unit covered by this specification is intended to be used for the lubrication of various types of automotive and heavy construction equipment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Accessories to be furnished with the lubricating and servicing unit (see 3.8).

c. Whether cartridge type hand grease guns and grease cartridges are to be furnished (see 3.8).

d. Whether technical publications are to be furnished (see 3.22).

e. Number of units required for preproduction testing (see 4.3.1).

f. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) form the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Air Force - 99

Preparing Activity:

Air Force - 82

Project No. 4930-F236

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-4541C(USAF)	LUBRICATING AND SERVICING UNIT, AIR OPERATED HIGH AND LOW

3a. NAME OF SUBMITTING ORGANIZATION PRESSURE DELIVERY

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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